

The Unified Hierarchical Reasoning Controller

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Abstract

This paper presents the Unified Hierarchical Reasoning Controller (UHRC), a novel framework designed to enhance the reasoning capabilities of artificial intelligence systems. The UHRC integrates hierarchical reasoning processes, allowing for more efficient and effective decision-making in complex environments. The results indicate that the UHRC can significantly improve the adaptability and robustness of AI systems, making it a promising approach for future research in artificial intelligence and machine learning.

1 Introduction

The autonomous control of complex nonlinear systems remains one of the most consequential open problems in control engineering (Iqbal et al. 2017; Z. Liu et al. 2015; Daoutidis, Megan, and W. Tang 2023). Systems such as unmanned aerial vehicles (UAVs), autonomous ground vehicles (UGVs), and high degree-of-freedom robotic manipulators are characterised by high dimensional state spaces, nonlinear inter-state couplings, actuator saturation, and significant parametric uncertainty (Nasser Sadati, Rahmani, and Saif 2015; Ameli and Anubi 2021; Iqbal et al. 2017; Slotine and W. Li 1991). The fundamental difficulties are that the dynamical regimes governing long horizon navigation and those governing fast stabilisation operate on incommensurable timescales, differ in their sensitivity to model error, and impose competing demands on the same actuator plant (Liang et al. 2018; Abdelmaksoud, Mailah, and Abdallah 2020). Designing a controller that simultaneously addresses long horizon planning and fast reactive stabilisation across these coupled dynamics is fundamentally a multiscale reasoning problem, and no single architecture has proven universally adequate across the competing demands of stability, adaptability, and computational tractability (Rojas, Yepes, and Garcia 2025; Goodwin, Graebe, and Salgado 2001).

Classical hierarchical control has long addressed this multi-scale challenge through explicit structural decomposition, a high-level supervisory layer generates strategic plans and setpoints while a low-level regulatory layer executes fast stabilisation and trajectory tracking (Singh and Titli 1978; Nasser Sadati, Rahmani, and Saif 2015; Scattolini and Colaneri 2007; Mesarovic, Macko, and Takahara 1970; Wilson 1979). The theoretical underpinnings of this separation trace to the multilevel systems framework of (Mesarovic, Macko, and Takahara 1970) and the hierarchical decomposition methods formalised in

(Singh and Titli 1978; Siljak 2013), which established that large scale optimisation problems can be rendered tractable by decomposing them into coordinated subproblems across levels. This architecture has achieved sustained success across aerospace, process control, and robotics, benefiting from well established stability theory, modular implementation, and interpretable failure modes (Sanders et al. 1998; X.-F. Wang, Jiang, and W.-H. Chen 2024; M. Liu, Tan, and Padois 2016; Bahnasawi, Al-Fuhaid, and M. Mahmoud 1990; Roberts and Becerra 2001). The quadrotor UAV exemplifies this paradigm directly, the natural timescale separation between slow translational dynamics and fast rotational dynamics motivates a cascaded structure in which outer loop position planners generate attitude references consumed by inner-loop attitude regulators (Hoffmann et al. 2007; Liang et al. 2018; S. Bouabdallah, Noth, and Siegwart 2004; Nagaty et al. 2013; Abdelhay and Zakriti 2019). Model predictive control has provided a principled realisation of the supervisory layer in such architectures, enabling constraint aware planning with formal stability guarantees at the coordination level (Scattolini and Colaneri 2007; X.-F. Wang, Jiang, and W.-H. Chen 2024; Torrente et al. 2021).

However, the fixed interface between layers introduces a structural information bottleneck that becomes consequential as task complexity grows (Kolchinsky, Tracey, and Wolpert 2019; S. Li et al. 2023; Abdelmaksoud, Mailah, and Abdallah 2020). The high-level planner typically transmits heavily compressed information ordinarily a kinematic waypoint sequence to the low-level regulator, discarding contextual information regarding obstacle proximity, actuator feasibility, and local geometric constraints that may be critical for downstream control quality (Zhu, Ferrari-Trecate, and Geroliminis 2024; Z. Zhou et al. 2017; W. Liu, Zheng, and Cai 2013). Conversely, the low-level regulator executes reference trajectories without access to the global context that motivated them, rendering local disturbance responses globally suboptimal (Nguyen, Lowenberg, and Neild 2021; Daoutidis, W. Tang, and Allman 2019). Fixed rule based supervisory frameworks also become brittle when the system encounters scenarios not anticipated at design time (N. Sadati and Ramezani 2008; Dave et al. 2023). These limitations are not incidental but structural, they arise from the requirement that the planner and regulator communicate through a prespecified, fixed dimensional interface whose information content is determined at design time rather than adapted online to task demands (Kolchinsky, Tracey, and Wolpert 2019; Zhang, Lu, and Gao 2015).

Learning based approaches have emerged as a principled alternative. End-to-end reinforcement learning (RL) enables joint optimisation of planning and regulation without manual interface specification (AlMahamid and Grolinger 2021; Levine et al. 2016; Kiran et al. 2022; Shakya, Pillai, and Chakrabarty 2023). Behavioural cloning provides a supervised alternative that sidesteps the credit assignment challenges of RL by learning directly from expert demonstrations, and has seen substantial success in autonomous driving and manipulation (Levine et al. 2016; Le Mero et al. 2022; Ly and Akhloufi 2021; Block, Foster, and Misra 2024). However, flat end-to-end policies are notoriously sample inefficient, sensitive to reward shaping, and do not naturally maintain the multiscale temporal structure that long horizon UAV navigation demands, the policy allocates identical computational depth regardless of whether the task requires strategic deliberation or reactive stabilisation (AlMahamid and Grolinger 2021; Agyei, Sarhadi, and Polani 2025). Deep reinforcement learning approaches applied directly to quadrotor control have demonstrated impressive agility in simulation (Trad et al. 2024; Saviolo and Loianno 2023), but the sim-to-real gap, sample complexity, and absence of formal stability guarantees remain open challenges (Dai et al. 2021; Agyei, Sarhadi, and Polani 2025). Hybrid neurosymbolic and

fuzzy integrated controllers partially address this brittleness by pairing learned components with structured decomposition (H. H. Tang and Ahmad 2024; T. A. Mahmoud et al. 2024; Tran et al. 2021; Baheri and Alm 2025), yet the supervisory-regulatory interface and its dimensionality remain manually prescribed. The boundary between planning and regulation is fixed by the designer, not learned from data (Borri, Pola, and Benedetto 2019; Yu et al. 2023).

Recent advances in neural reasoning architectures offer a substantially different approach to this problem. Test time scaling methods that iterate computations in latent space (Geiping et al. 2025; H. Li et al. 2025) and recurrent architectures that adapt computational depth to task difficulty (Geiping et al. 2025; Z.-Z. Li et al. 2025) suggest that the depth of processing required for a given decision can be made emergent rather than fixed. The Hierarchical Reasoning Model (HRM), introduced by (G. Wang et al. 2025), is a two tier recurrent architecture comprising a high-level module responsible for slow, abstract planning and a low-level module handling rapid, detailed computation, coupled through a shared carry state that enables adaptive allocation of computational depth. Trained on as few as 1,000 examples without pre-training or chain-of-thought supervision, the HRM achieves near perfect accuracy on complex Sudoku problems and optimal pathfinding in large maze tasks on which state-of-the-art large language model approaches fail entirely (G. Wang et al. 2025; J. Huang and Chang 2023; Feng et al. 2025). Crucially, the hierarchical role specialisation between modules emerges through training rather than being architecturally prescribed, mirroring the representational dimensionality hierarchy observed across cortical regions in the mammalian brain (Badre and Nee 2018; Huntenburg, Bazin, and Margulies 2018; Murray et al. 2014), where higher order areas operate over longer intrinsic timescales and higher dimensional representations than primary sensory regions. Building on this, (Jin et al. 2025) demonstrated that decoupling planning from execution in a hierarchical framework yields measurable gains in multi-step reasoning tasks, further motivating the application of this design principle to closed loop control. The hierarchical neuro-symbolic decision transformer of (Baheri and Alm 2025) similarly demonstrates that coupling a symbolic planner with a learned low-level policy through a structured interface outperforms both purely symbolic and purely neural baselines on sequential decision tasks, providing additional evidence that the hierarchical planning-execution separation confers a genuine computational advantage beyond what either component achieves in isolation.

Nevertheless, all demonstrated HRM applications to date have been confined to discrete, turn based domains. Whether the same unified reasoning mechanism can function as a closed loop controller for continuous, real time nonlinear dynamical systems where control outputs must be physically continuous, observations arrive as dense sensory streams, and temporal context must persist across thousands of control cycles remains entirely unexplored in the literature. Extending HRMs to such settings requires nontrivial architectural adaptations, discrete token outputs must be replaced by continuous commands, and the supervised training signal must be derived from a physically meaningful expert policy rather than symbolic ground truth labels. Localisation and mapping, which are prerequisite for any meaningful planning computation, must be supplied by an onboard sensor (Gupta and Fernando 2022; Milijas et al. 2021; Munguia, Trujillo, and Grau 2025; W. Chen et al. 2022) whose outputs form part of the continuous observation stream consumed by the controller.

This paper addresses that gap by proposing and evaluating the *Unified Hierarchical Reasoning Controller* (UHRC) for the autonomous navigation of a six-degree-of-freedom

quadrotor UAV, constituting the first systematic application of the HRM architecture to a continuous nonlinear closed loop control system. The UHRC adapts the HRM’s two-level reasoning structure into a recurrent transformer controller trained via behavioural cloning (Levine et al. 2016; Le Mero et al. 2022; Block, Foster, and Misra 2024) on expert trajectory data generated by an A* path planner (LaValle 2006; Qin et al. 2023) across 500 simulated episodes. The architecture is evaluated against a classical hierarchical baseline incorporating SLAM-based localisation (Gupta and Fernando 2022; Milijas et al. 2021), A* path planning, and cascaded PID control (S. Bouabdallah, Noth, and Siegwart 2004; J. Ren et al. 2016; Abdelhay and Zakriti 2019; Lopez-Sanchez and Moreno-Valenzuela 2023) across five structured test cases of increasing environmental complexity.

The primary contributions of this work are as follows:

1. The first empirical demonstration of an HRM-based architecture operating as a closed loop controller for a continuous nonlinear dynamic system, establishing the functional feasibility of unified hierarchical reasoning in the control domain.
2. A UHRC architecture incorporating dual-stream fusion of kinematic and LiDAR observations, a rolling carry buffer for persistent temporal context, Rotary Positional Encoding (RoPE) over the carry sequence, and a gated subgoal injection mechanism coupling H-module planning to L-module computation.
3. A behavioural cloning training pipeline employing a versatile episode distribution, A*-generated expert demonstrations with injected perturbations and a structured loss combining weighted action imitation, subgoal supervision, and alignment.
4. A rigorous comparative empirical evaluation quantifying the performance robustness trade off between unified learned hierarchy and classical explicit decomposition across open field transit, constrained corridor navigation, gap threading, dynamic obstacle avoidance, and parametric model uncertainty.

2 Methodology

This section details the UHRC architecture, the classical hierarchical baseline against which it is evaluated, the expert data generation procedure, the behavioural cloning training pipeline, and the comparative evaluation protocol. All controllers share an identical 6-DOF quadrotor plant, described first.

2.1 Quadrotor Dynamics Model

The benchmark plant is a rigid-body quadrotor modelled in the East-North-Up (ENU) inertial frame $\mathcal{I}$ with body frame $\mathcal{B}$ fixed to the vehicle centre of mass. The state vector is

$$\mathbf{x} = [\mathbf{r}_I^\top, \mathbf{v}_I^\top, \mathbf{q}_{BI}^\top, \boldsymbol{\omega}_B^\top, \boldsymbol{\Omega}^\top]^\top \in \mathbb{R}^{17}, \quad (1)$$

where $\mathbf{r}_I \in \mathbb{R}^3$ and $\mathbf{v}_I \in \mathbb{R}^3$ are the inertial position and velocity, $\mathbf{q}_{BI} \in \mathbb{R}^4$ is the unit quaternion representing the rotation from $\mathcal{I}$ to $\mathcal{B}$ satisfying $\|\mathbf{q}_{BI}\| = 1$, $\boldsymbol{\omega}_B \in \mathbb{R}^3$ is the angular velocity expressed in $\mathcal{B}$, and $\boldsymbol{\Omega} \in \mathbb{R}^4$ contains the four rotor speeds.

The control input is the body-frame wrench

$$\mathbf{u} = [T, \tau_x, \tau_y, \tau_z]^\top \in \mathbb{R}^4, \quad (2)$$

comprising the net collective thrust T along the body $+z$ axis and roll, pitch, and yaw torques τ_x , τ_y , τ_z respectively. This formulation bypasses explicit rotor speed dynamics and assumes ideal wrench actuation, which is appropriate for the simulation study conducted here (Hoffmann et al. 2007).

The translational dynamics are governed by Newton's second law in the inertial frame:

$$m\dot{\mathbf{v}}_I = mg + R_{IB} \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix} + C_d^{\text{lin}} \mathbf{v}_I, \quad (3)$$

where m is the vehicle mass, g is gravitational acceleration, $R_{IB} = R_{BI}^\top$ is the rotation matrix from $\mathcal{B}$ to $\mathcal{I}$ derived from $\mathbf{q}_{BI}$, and $C_d^{\text{lin}} = \text{diag}(c_x, c_y, c_z)$ is the linear aerodynamic drag matrix. The rotational dynamics follow the Newton-Euler equations in the body frame:

$$J\dot{\boldsymbol{\omega}}_B = \boldsymbol{\tau}_B - \boldsymbol{\omega}_B \times J\boldsymbol{\omega}_B - C_d^{\text{rot}} \boldsymbol{\omega}_B, \quad (4)$$

where $J = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ is the diagonal inertia matrix, $\boldsymbol{\tau}_B = [\tau_x, \tau_y, \tau_z]^\top$, and $C_d^{\text{rot}} = \text{diag}(d_x, d_y, d_z)$ is the rotational damping matrix. Attitude kinematics are propagated via the unit quaternion differential equation

$$\dot{\mathbf{q}}_{BI} = \frac{1}{2} (\mathbf{q}_{BI}) \boldsymbol{\omega}_B, \quad (5)$$

where $(\mathbf{q}) \in \mathbb{R}^{4 \times 3}$ is the standard quaternion kinematic matrix (Shstry, Kothari, and Abhishek 2018). The system is integrated using a fourth-order Runge-Kutta scheme at a fixed timestep of $\Delta t = 0.01$ s, with quaternion renormalisation applied after each step. Physical parameters are listed in Table 1.

Table 1: Quadrotor physical parameters.

Parameter	Symbol	Value	Units
Mass	m	1.0	kg
Inertia (roll)	I_{xx}	0.008	kg m ²
Inertia (pitch)	I_{yy}	0.008	kg m ²
Inertia (yaw)	I_{zz}	0.014	kg m ²
Arm length	l	0.17	m
Thrust coefficient	k_T	1.9×10^{-6}	N/(rad/s) ²
Torque coefficient	k_M	2.6×10^{-8}	N·m/(rad/s) ²
Max rotor speed	Ω_{max}	18,000	rpm
Linear drag	$\text{diag}(C_d^{\text{lin}})$	[0.10, 0.10, 0.18]	N/(m/s)
Rotational drag	$\text{diag}(C_d^{\text{rot}})$	[0.02, 0.02, 0.02]	N·m/(rad/s)

2.2 Classical Hierarchical Baseline

The baseline is a four-stage classical hierarchical controller implementing the full supervisory-regulatory decomposition described in the literature (Sanders et al. 1998; Scattolini and Colaneri 2007). It serves as a representative instance of the architecture class the UHRC is designed to replace.

An Inertial Navigation System Extended Kalman Filter (INSEKF) fuses GPS position and Attitude and Heading Reference System (AHRS) orientation estimates at 10 Hz to

produce a pose estimate $[\hat{\mathbf{r}}_I, \hat{\mathbf{q}}_{BI}]$. LiDAR odometry is provided by an Iterative Closest Point (ICP) registration pipeline operating on 2D point cloud keyframes, accumulated into a pose graph optimised via Levenberg-Marquardt at fixed intervals of 500 nodes (Gupta and Fernando 2022; Milijas et al. 2021).

A collision-free path is computed by the A* algorithm on a 2D occupancy grid constructed by projecting LiDAR returns onto the ground plane at 2.0 m cell resolution. The grid spans the drone’s estimated flight level within a height band of ± 2.0 m. Path feasibility is guaranteed if a collision-free grid path exists (C. Zhou, B. Huang, and Fränti 2021); otherwise the controller holds the previously valid path.

A pure pursuit tracker with a fixed lookahead distance of $L_d = 0.8$ m translates the planned waypoint sequence into position setpoints at 10 Hz (Y. Huang et al. 2020).

A cascaded PID structure realises the position setpoints. The outer position loop maps horizontal position errors to desired roll and pitch references $[\phi_d, \theta_d]$ via the standard small-angle coupling $\theta_d = a_x^{\text{des}}/g$, $\phi_d = -a_y^{\text{des}}/g$. The inner attitude loop generates body wrench commands from attitude and altitude errors using derivative-filtered PID controllers on each axis (T. Ren and Jebelli 2025; Samir Bouabdallah 2007).

2.3 UHRC Architecture

The UHRC is a two-level recurrent transformer that adapts the HRM architecture (G. Wang et al. 2025) to continuous closed-loop control. At each timestep, a high-level module H generates a 3-dimensional velocity subgoal encoding strategic intent, and a low-level module L translates that intent together with a rich sensory context into a 4-dimensional body wrench command. Both modules maintain persistent memory through a rolling carry buffer that replaces the HRM’s per-problem hidden state, enabling temporal reasoning across the full episode horizon.

2.3.1 Observation Representation

All observations are expressed in the body frame $\mathcal{B}$. This choice renders the observation invariant to global heading and position, reducing the domain of the mapping the controller must learn: an identical avoidance manoeuvre is appropriate regardless of the vehicle’s absolute world-frame orientation Liang et al. 2018. The 45-dimensional observation vector is

$$\mathbf{o} = \left[\underbrace{\mathbf{v}_B^\top}_{\times 3}, \underbrace{\boldsymbol{\omega}_B^\top}_{\times 3}, \underbrace{\mathbf{g}_B^\top}_{\times 3}, \underbrace{\Delta \mathbf{r}_B^\top}_{\times 3}, \underbrace{d_{\text{norm}}}_{\times 1}, \underbrace{\mathbf{r}_{\text{lidar}}^\top}_{\times 32} \right]^\top \in \mathbb{R}^{45}, \quad (6)$$

where $\mathbf{v}_B = R_{BI}\mathbf{v}_I$ is the body-frame translational velocity, $\boldsymbol{\omega}_B$ is the angular velocity, $\mathbf{g}_B = R_{BI}[0, 0, -g]^\top$ encodes attitude implicitly by projecting the gravity vector into $\mathcal{B}$, $\Delta \mathbf{r}_B = R_{BI}(\mathbf{r}_{\text{goal}} - \mathbf{r}_I)$ is the goal displacement clipped to a maximum range $r_{\text{max}}^{\text{goal}} = 20$ m, $d_{\text{norm}} = \min(\|\Delta \mathbf{r}_B\|, 20)/20$ is the normalised goal distance, and $\mathbf{r}_{\text{lidar}} \in [0, 1]^{32}$ is a 32-ray planar LiDAR scan with field of view π rad and maximum range 5.0 m, normalised to the unit interval.

The separate scalar d_{norm} is included deliberately, without it, the goal displacement vector $\Delta \mathbf{r}_B$ loses magnitude information as the drone approaches the goal, causing the controller to overshoot. At runtime, the observation is standardised by precomputed dataset statistics before being passed to the model.

2.3.2 Dual-Stream Input Encoder

The 13 scalar kinematic features $[\mathbf{v}_B^\top, \boldsymbol{\omega}_B^\top, \mathbf{g}_B^\top, \Delta \mathbf{r}_B^\top, d_{\text{norm}}]$ and the 32-ray LiDAR scan are processed by separate encoder branches before fusion.

A two-layer MLP first projects the 13-dimensional input to $\mathbb{R}^{h/2}$ through a scaled hyperbolic tangent:

$$\mathbf{s} = W_2 \tanh(\sqrt{H} \cdot W_1 \mathbf{o}_{\text{kin}} + \mathbf{b}_1), \quad (7)$$

where $h = 128$ is the model hidden dimension. The scaling factor $\sqrt{H}$ expands the effective dynamic range of kinematic inputs into the regime where $\tanh$ provides informative gradients, preventing early saturation during training.

The 32 normalised range returns are processed as a 1D signal by a two-stage convolutional encoder:

$$\mathbf{l} = W_{\text{proj}} \text{Flatten}(\sigma(\text{Conv}_{k=3}^{16}(\sigma(\text{Conv}_{k=5}^{16}(\mathbf{o}_{\text{lidar}}))))), \quad (8)$$

where σ denotes the SiLU activation, Conv_k^c denotes a 1D convolution with c output channels and kernel size k (same padding), and $W_{\text{proj}} \in \mathbb{R}^{(h/2) \times 512}$ projects the flattened $16 \times 32 = 512$ feature map to $\mathbb{R}^{h/2}$.

The two $h/2$ -dimensional branch outputs are concatenated and linearly projected to a single embedding $\mathbf{e}_t = W_f[\mathbf{s}; \mathbf{l}] \in \mathbb{R}^h$ representing the current observation at timestep t .

2.3.3 Transformer Reasoning Blocks

Both the H -module and L -module are composed of stacked `UHRCBlock` layers, each implementing a standard pre-normalised transformer encoder block:

$$\mathbf{z}' = \text{RMSNorm}(\mathbf{z} + \text{MHA}(\mathbf{z})), \quad (9)$$

$$\mathbf{z}'' = \text{RMSNorm}(\mathbf{z}' + \text{SwiGLU}(\mathbf{z}')), \quad (10)$$

where $\text{MHA}(\cdot)$ is non-causal multi-head self-attention with $n_h = 4$ heads and head dimension $d_h = h/n_h = 32$, and $\text{SwiGLU}(\cdot)$ is the gated feed-forward network

$$\text{SwiGLU}(\mathbf{z}) = W_{\downarrow}(\text{SiLU}(W_g \mathbf{z}) \odot W_u \mathbf{z}), \quad (11)$$

with intermediate dimension $d_{\text{ff}} = 512$. Non-causal attention is appropriate here because the carry represents a temporal history rather than an autoregressive prediction task, each position can attend to all other positions, including more recent ones.

Rotary Positional Encoding (RoPE) is applied independently to the query and key matrices within each attention head, encoding the relative age of each position in the carry buffer. For a sequence of length T and head dimension d_h , the rotary frequencies are

$$\theta_i = \frac{1}{10000^{2i/d_h}}, \quad i = 0, 1, \dots, d_h/2 - 1, \quad (12)$$

with cosine and sine tables precomputed up to the maximum sequence length of $T_{\text{max}} = \text{carry_len} + 1 = 17$. RoPE is preferred over learned absolute positional embeddings because it generalises to sequence positions not seen during training (G. Wang et al. 2025).

2.3.4 Persistent Temporal Carry Buffer

In the original HRM, hidden states $\mathbf{z}_H$ and $\mathbf{z}_L$ are re-initialised for each input problem. In closed-loop control, the controller must maintain situational awareness across successive timesteps, tracking velocity history, recognising developing obstacle geometries, and anticipating trajectory evolution. The UHRC therefore treats the hidden states as persistent rolling buffers:

$$\mathbf{z}_H, \mathbf{z}_L \in \mathbb{R}^{B \times C \times h}, \quad (13)$$

where B is the batch size (1 at inference), $C = 16$ is the carry length, and $h = 128$ is the hidden dimension. At each timestep, the new observation embedding $\mathbf{e}_t$ is appended to the buffer and the oldest entry is discarded:

$$\text{roll_carry}(\mathbf{z}, \mathbf{e}_t) = [\mathbf{z}_{:,1:C,:}; \mathbf{e}_t^\top] \in \mathbb{R}^{B \times C \times h}, \quad (14)$$

providing a fixed-length sliding window of temporal context without unbounded memory growth. The carry therefore functions as a learned recurrent state that compresses the episode’s observational history into a fixed-size representation, structurally analogous to the reasoning context in recurrent language models.

Carry tokens appended to the buffer are always gradient-detached (`DETACH_CARRY=True`). This truncates backpropagation through the carry boundary, preventing gradient chains from extending beyond the training sequence window of length $T_{\text{seq}} = 8$ steps. Empirical evaluation confirmed that this convention does not impair the quality of learned temporal representations within the window, consistent with findings in the HRM literature (G. Wang et al. 2025). At the start of each episode, both buffers are initialised to learned constant vectors $\mathbf{h}_H^{\text{init}}$ and $\mathbf{h}_L^{\text{init}}$ tiled to shape $[1, C, h]$.

2.3.5 H-L Mutual Refinement and One-Step Gradient Approximation

The H - and L -modules interact through $H_{\text{cyc}} = 2$ outer cycles each containing $L_{\text{cyc}} = 2$ inner L -module passes. At each step, the full sequence $\mathbf{z}_H^{(t)} \in \mathbb{R}^{B \times 17 \times h}$ is formed by concatenating the $C = 16$ carry tokens with the current observation embedding:

$$\mathbf{z}_H^{(t)} = [\mathbf{Z}_H; \mathbf{e}_t^\top], \quad \mathbf{z}_L^{(t)} = [\mathbf{Z}_L; \mathbf{e}_t^\top]. \quad (15)$$

An observation re-injection tensor $\mathbf{I}_{\text{obs}} \in \mathbb{R}^{B \times 17 \times h}$ is constructed with zeros at carry positions and $\mathbf{e}_t$ at position 16, ensuring the raw observation is available as an additive bias at every L -module call — a critical structural correspondence to the HRM’s input injection mechanism (G. Wang et al. 2025).

To bound the gradient graph depth and maintain training stability, the UHRC adopts the one-step gradient approximation from the HRM: all but the final module call in the refinement cycle are computed under `torch.no_grad()`, and only a single L -pass followed by a single H -pass contribute to the gradient. With $H_{\text{cyc}} = 2$ and $L_{\text{cyc}} = 2$, the execution schedule is:

- i. **No-grad:** $L \rightarrow L \rightarrow H \rightarrow L$ (3 L -calls, 1 H -call)
- ii. **Gradient:** $L \rightarrow H$ (1 L -call, 1 H -call)

The effective gradient path traverses $2 + 2 = 4$ transformer layers in total, roughly $3.5\times$ shallower than the full 14-layer computational graph, preventing gradient vanishing while

retaining the representational benefit of multiple refinement rounds. The L -module update at each call is

$$\mathbf{z}_L \leftarrow \mathcal{L}(\mathbf{z}_L + \mathbf{z}_H + \mathbf{I}_{\text{obs}}), \quad (16)$$

and the H -module update is

$$\mathbf{z}_H \leftarrow \mathcal{H}(\mathbf{z}_H + \mathbf{z}_L), \quad (17)$$

where $\mathcal{L}(\cdot)$ and $\mathcal{H}(\cdot)$ denote the respective stacks of UHRCBlock layers.

2.3.6 Subgoal Generation, Gated Injection, and Action Output

Following the final H -module update, the terminal hidden state $\mathbf{z}_H[:, 16, :] \in \mathbb{R}^{B \times h}$ (the most recent position in the carry sequence) is decoded by a two-layer MLP planning head into a 3-dimensional body-frame velocity subgoal:

$$\hat{\mathbf{v}}_{\text{sub}} = W_2 \text{SiLU}(W_1 \mathbf{z}_H[:, 16, :]) \in \mathbb{R}^{B \times 3}. \quad (18)$$

This subgoal encodes the H -module’s strategic intent as a desired velocity vector $[v_x, v_y, v_z]$ in the body frame, providing a physically interpretable interface between the two modules.

The subgoal is injected into the L -module through a gated additive mechanism designed to modulate the reactive representation without overriding it:

$$\mathbf{g} = \sigma(W_g \hat{\mathbf{v}}_{\text{sub}} + \mathbf{b}_g) \in (0, 1)^{B \times h}, \quad (19)$$

$$\mathbf{c} = (W_p \hat{\mathbf{v}}_{\text{sub}}) \odot \mathbf{g} \in \mathbb{R}^{B \times h}, \quad (20)$$

where σ is the element-wise sigmoid. The gate bias is initialised to $\mathbf{b}_g = +2.0$, so that $\sigma(+2) \approx 0.88$ at the start of training keeping the gate mostly open and allowing the subgoal signal to propagate freely before the planning head has converged. The gated context is broadcast across the carry sequence and the L -module performs a final two-layer attention pass:

$$\mathbf{z}_L^{\text{final}} = \mathcal{L}(\mathbf{z}_L + \mathbf{c} \cdot \mathbf{1}_{17}^\top), \quad (21)$$

from which the body-frame wrench action is decoded from the terminal position:

$$\hat{\mathbf{u}} = W_a \mathbf{z}_L^{\text{final}}[:, 16, :] = [\hat{F}_z, \hat{\tau}_x, \hat{\tau}_y, \hat{\tau}_z]^\top \in \mathbb{R}^4. \quad (22)$$

The action head weight is initialised with standard deviation 10^{-2} and the thrust bias is set to $\hat{F}_z^{(0)} = 9.81 \text{ N}$ (hover thrust for a 1 kg vehicle), ensuring the network produces a stable hover at initialisation before any learning has occurred. The full architectural configuration is summarised in Table 2.

2.4 Expert Data Generation

Training data are generated by rolling out an expert policy comprising an A* path planner, a pure-pursuit tracker, and the cascaded PID attitude controller on 500 successful simulated episodes at $\Delta t = 0.01 \text{ s}$ per step (Levine et al. 2016; Le Mero et al. 2022).

At the start of each episode, a 2D occupancy grid is constructed at a resolution of $r_{\text{grid}} = 0.25 \text{ m}$ per cell, spanning the convex hull of the start position, goal position, and all obstacle extents with a 2.0 m margin. Circular obstacles are inflated by $r_{\text{inf}} = 0.25 \text{ m}$ to account for vehicle body radius. The A* algorithm searches an 8-connected neighbourhood using the octile heuristic, guaranteeing the shortest collision-free path if one exists (C.

Table 2: UHRC fixed architectural configuration.

Parameter	Symbol	Value
State dimension	d_s	45
LiDAR dimension	d_{lidar}	32
Action dimension	d_a	4
Subgoal dimension	d_{sub}	3
Hidden size	h	128
Carry length	C	16
Attention heads	n_h	4
Head dimension	d_h	32
FFN intermediate	d_{ff}	512
H-module layers	N_H	2
L-module layers	N_L	2
H-level cycles	H_{cyc}	2
L-level cycles	L_{cyc}	2
RoPE base frequency	—	10,000
Hover thrust bias	$F_z^{(0)}$	9.81 N

Zhou, B. Huang, and Fränti 2021). Episodes for which no path exists are discarded before simulation begins.

The planned path is smoothed by three passes of a cubic averaging filter and tracked by a pure-pursuit controller with lookahead $L_d = 1.0$ m and maximum speed $V_{\text{max}} = 2.0$ m/s. Speed is linearly ramped down to 0.3 m/s within 2 m of the goal. Upon entering a 0.1 m goal acceptance radius, the tracker issues a zero velocity command for 150 additional steps (1.5 s), demonstrating the hover termination behaviour to the model.

To promote policy generalisation beyond the default $+x$ -direction transit, six episode categories are sampled according to the distribution in Table 3.

Table 3: Episode type distribution for expert data generation.

Type	Proportion	Description
Normal	35%	Long-range $+x$ transit with 4–8 obstacles
Omnidirectional	20%	Random start/goal in full arena, any heading
No obstacle	15%	Long-range transit, obstacle-free
Close	10%	Short-range (1–5 m), any direction, 0–2 obstacles
Recovery	10%	Start beyond or beside goal; teaches re-acquisition
Tight gap	10%	Symmetric wall pair with 0.5–0.8 m half-gap

The omnidirectional episodes are critical for eliminating the directional bias intrinsic to the dominant $+x$ -transit training distribution: without them, the learned policy fails to navigate in the $-x$ or $\pm y$ directions.

Every 50 steps with probability 0.3, the current state receives a Gaussian velocity perturbation ($\sigma = 0.2$ m/s) and a small random yaw offset ($|\delta\psi| \leq 0.15$ rad). This DAgger-style augmentation (Levine et al. 2016) exposes the expert to a wider distribution of states and prevents the compounding error pathology characteristic of pure behavioural cloning.

Episodes are included only if: (i) the drone reached the goal without collision; (ii) the goal distance decreased monotonically over coarse 100-step windows; and (iii) the average velocity subgoal magnitude during the final hover phase was below 0.3 m/s. These criteria

filter episodes in which the expert made insufficient progress or failed to demonstrate a clean stopping behaviour.

2.5 Behavioural Cloning Training

2.5.1 Dataset Construction

The **ControlDataset** constructs sequence windows of length $T_{\text{seq}} = 8$ timesteps from the stored episode data. Windows never cross episode boundaries; windows that begin within the first $T_{\text{seq}} - 1$ steps of an episode are left-padded with zeros. The 45-dimensional observations are z -scored using dataset-wide mean and standard deviation statistics. Actions and velocity subgoals are retained in their raw physical units (N and m/s respectively) so that the magnitude of control effort and velocity urgency are preserved as supervisory signals. The dataset is split into training and validation subsets by episode, with 10% of episodes reserved for validation.

2.5.2 Loss Function

Three loss terms are combined:

$$\mathcal{L} = \mathcal{L}_{\text{act}} + \alpha_s \mathcal{L}_{\text{sub}} + \alpha_t \mathcal{L}_{\text{trk}}, \quad (23)$$

with $\alpha_s = 0.1$ and $\alpha_t = 0.5$.

The four action components $[F_z, \tau_x, \tau_y, \tau_z]$ have very different variances in the training data: thrust is tightly concentrated near the hover value $F_z^{\text{hover}} = 9.81$ N, while the torque components span a much smaller absolute range but carry proportionally more information per unit deviation. A per-component weighted MSE is used to compensate:

$$\mathcal{L}_{\text{act}} = \left\langle \left(\hat{\mathbf{u}} - \mathbf{u}^* \right)^\top \text{diag}(\mathbf{w}_a) \left(\hat{\mathbf{u}} - \mathbf{u}^* \right) \right\rangle, \quad (24)$$

where $\langle \cdot \rangle$ denotes the mean over the batch and $\mathbf{w}_a = [0.68, 71.4, 69.0, 50.0] / \bar{w}_a$ are the normalised component weights.

An analogous weighted MSE on the 3-dimensional velocity subgoal:

$$\mathcal{L}_{\text{sub}} = \left\langle \left(\hat{\mathbf{v}}_{\text{sub}} - \mathbf{v}_{\text{sub}}^* \right)^\top \text{diag}(\mathbf{w}_s) \left(\hat{\mathbf{v}}_{\text{sub}} - \mathbf{v}_{\text{sub}}^* \right) \right\rangle, \quad (25)$$

with weights $\mathbf{w}_s = [1.5, 1.5, 0.1] / \bar{w}_s$. The vertical component v_z is down-weighted because episodes are conducted at constant altitude, making the vertical subgoal largely uninformative.

The H-module subgoal and L-module torques must be physically consistent: pitch torque τ_y produces forward acceleration that realises v_x , and roll torque τ_x produces lateral acceleration that realises v_y . The tracking loss penalises misalignment between the horizontal subgoal direction $[v_x, v_y]^\top$ and the corresponding torque vector $[\tau_y, \tau_x]^\top$:

$$\mathcal{L}_{\text{trk}} = \left\langle \max \left(0, -\frac{[\hat{v}_x, \hat{v}_y]}{\|[\hat{v}_x, \hat{v}_y]\| + \epsilon} \cdot \frac{[\hat{\tau}_y, \hat{\tau}_x]}{\|[\hat{\tau}_y, \hat{\tau}_x]\| + \epsilon} \right) \right\rangle. \quad (26)$$

This term acts as a soft architectural consistency constraint, encouraging the inter-module coupling to reflect the correct body-frame dynamics.

2.5.3 Optimiser and Schedule

The model is trained with AdamW Levine et al. 2016 using the configuration in Table 4. A linear warmup phase over the first 5 epochs raises the learning rate from zero to its peak value, after which cosine annealing decays it to approximately zero over the remaining 15 epochs. Gradients are clipped to a global ℓ_2 norm of 1.0 at each step. Automatic mixed precision (AMP) with a gradient scaler is employed, reducing memory consumption to accommodate a batch size of 512.

Table 4: Behavioural cloning training configuration.

Parameter	Symbol	Value
Sequence length	T_{seq}	8
Batch size	B	512
Peak learning rate	η	2×10^{-4}
Weight decay	λ	3×10^{-3}
Gradient clip	—	1.0
Warmup epochs	—	5
Total epochs	—	20
Validation split	—	10% (by episode)
Subgoal loss weight	α_s	0.1
Tracking loss weight	α_t	0.5
Action weights	$\mathbf{w}_a$	[0.68, 71.4, 69.0, 50.0]
Subgoal weights	$\mathbf{w}_s$	[1.5, 1.5, 0.1]

2.6 Comparative Evaluation Protocol

Both controllers are evaluated across five structured test cases (TC01–TC05) of increasing environmental complexity. To ensure a fair comparison, both controllers receive ideal state feedback from the same underlying simulation state; neither suffers from sensor noise or state estimation error. Start positions, goal positions, and obstacle configurations are seeded identically for both controllers. Each test case is repeated for $N = 30$ randomised Monte Carlo trials and metrics are averaged over successful and failed trials.

TC01 – Open Transit. Straight-line goal-seeking in an obstacle-free environment, establishing baseline trajectory tracking performance.

TC02 – Symmetric Corridor. Six obstacles arranged symmetrically around the nominal flight path, testing multi-obstacle avoidance and path quality under spatial constraint.

TC03 – Gap Navigation. Two obstacles forming a 3 m gap directly in the nominal flight path, assessing control precision and gap-threading capability.

TC04 – Dynamic Obstacle Avoidance. Obstacles are excluded from the controller’s sensor data until $t \geq 3.0$ s, then revealed, requiring mid-flight reactive replanning. Both controllers receive identical sensor data at each timestep.

TC05 – Parametric Robustness. The TC02 obstacle configuration is retained but the plant dynamics are perturbed: mass $\pm 20\%$, diagonal inertia $\pm 30\%$, drag coefficients $\pm 50\%$, and thrust coefficient $k_T \pm 15\%$. This characterises each controller’s robustness margin under distributional shift.

The primary performance metrics are root mean square error (RMSE) of the Euclidean position error, integral absolute error (IAE), integral time-weighted absolute error (ITAE),

path length ratio (PLR = actual path length / straight-line distance), episode success rate (goal reached within 0.5 m without collision), and collision rate.

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